

# Scorecard Update

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## Overview

This markdown shows the data logging process of each round, and shows performance trends and metric averages

## Set-Up Environment

### Attach Packages

```
library(golf)
library(tidyverse)
library(DBI)
library(duckdb)
```

## Record New Scorecard

### Input the Scores Data

```
# required inputs to fill out the scorecard

round_course <- 'Dell Urich'
round_date <- '2026-08-16'
round_teens <- 'combo'

hole_scores <- c(5, 5, 4, 2, 4, 5, 4, 6, 4,
                4, 4, 5, 5, 5, 4, 4, 4, 4)

FIRs <- c(1, 0, 1, 0, 0, 1, rep(0, 3),
          0, 0, 1, 0, 0, 1, 1, 0, 0)

GIRs <- c(0, 0, 1, 1, 1, 0, 0, 0, 0,
          0, 0, 0, 0, 0, 1, 1, 0, 1)

putts_rec <- c(2, 2, 1, 0, 3, 1, 1, 2, 2,
               1, 2, 2, 1, 2, 2, 2, 2, 2)

chips_rec <- c(1, 2, 0, 1, 0, 1, 1, 2, 1,
```

```

        1, 1, 1, 2, 1, 0, 0, 1, 0)

penalties_rec <- c(rep(0, 18))

tee_clubs <- c('D', '8', 'D', 'D', '9', 'D', 'D', '3W', '7',
              'D', '9', 'D', 'D', 'D', 'D', '4', 'PW', 'D')

index <- 9.9

```

## Specify Course and Tees

```

# get the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
  Card <- golf::get_course(course = round_course, date = round_date, tees = round_tees)
}

```

## Get Scoring Metrics

```

# fill the scorecard for the new round, ensuring hole-by-hole scores are filled
if ( length(hole_scores) > 0 ) {
  Card <- golf::log_score(Scorecard = Card,
                        hole_by_hole = hole_scores,
                        name = 'Erik Larsen',
                        index = index,
                        FIR = FIRs,
                        GIR = GIRs,
                        putts = putts_rec,
                        chips = chips_rec,
                        penalties = penalties_rec,
                        tee_club = tee_clubs)
}

```

## Update the db

### Update the Players Table

```

con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'players' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC"))
    dplyr::distinct(date) |>
    unlist() %>%
    lubridate::as_date(.) |>
    as.character() < round_date &

    length(hole_scores) > 0

```

```

    ) {

DBI::dbAppendTable(conn = con,
  name = 'players',
  value = players <- Card |>
    dplyr::select(player_id, player_name, GHIN, index, date) |>
    dplyr::distinct() |>
    dplyr::rename(handicap_index = index) |>
    dplyr::mutate(date = as.character(date))
  )
}

```

## Update the Courses Table

```

con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'courses' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC")) |>
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
DBI::dbAppendTable(conn = con,
  name = 'courses',
  value = course <- Card |>
    dplyr::select(course, tees, to_par, slope, course_rating, hole, yds, par, hole_handicap) |>
    dplyr::distinct() |>
    dplyr::rename(course_name = course) |>
    dplyr::mutate(hole = gsub(hole,
      pattern = 'hole_',
      replacement = '') |> as.numeric())
  )
}

```

## Update the Rounds Table

```

con <- golf::get_db_connection()

# if the logged data is newer than the last logged round in the 'rounds' table,
# update the 'rounds' table

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM rounds ORDER BY date DESC")) |>
  dplyr::distinct(date) |>
  unlist() %>%

```

```

lubridate::as_date(.) |>
as.character() < round_date &

length(hole_scores) > 0
) {
DBI::dbAppendTable(conn = con,
  name = 'rounds',
  value = Card |>
    dplyr::select(-to_par, -slope, -course_rating, -yds, -par) |>
    dplyr::distinct() |>
    dplyr::rename(handicap_index = index) |>
    dplyr::rename(course_name = course) |>
    dplyr::mutate(date = as.character(date)) |>
    dplyr::mutate(hole = gsub(hole,
      pattern = 'hole_',
      replacement = '') |> as.numeric()) |>
    dplyr::group_by(course_name, date) |>
    dplyr::arrange(hole) |>
    dplyr::ungroup() |>
    dplyr::relocate(c(tot_gross, tot_net), .after = IN_net) |>
    dplyr::relocate(course_handicap, .after = tees)
  )
}

```

## Get Club Metrics df

```

# create a dataframe, specifying the shape for tracked shots on each hole
# -> all strokes from off the green

club_metrics_df <- golf::get_tracked_shots_data_shape(round_date = round_date) |>
  dplyr::arrange(hole)

```

## Annotate Club Metrics

```

# manually annotate from Garmin Golf App log
club_choice <- c(
  'D', '6', 'SW',
  '8', 'SW', 'SW',
  'D', '3W', 'GW',
  'D', 'SW',
  '9',
  'D', '3W', '9', 'P',
  'D', 'GW', 'P',
  '3W', '7', 'SW', 'GW',
  '8', 'GW',

  'D', 'SW', 'PW',
  '9', 'LW',
  'D', 'SW', 'PW',
  'D', '5', 'GW', 'GW',

```

```

'D', 'PW', 'LW',
'D', 'SW',
'4', 'GW',
'PW', 'GW',
'D', 'GW'
)

dist_to_target <- c(
  275, 192, 42,
  161, 44, 35,
  275, 266, 43,
  275, 34,
  158,
  275, 253, 156, 8,
  275, 133, 7,
  255, 70, 74, 31,
  179, 22,

  275, 57, 11,
  152, 17,
  275, 94, 31,
  275, 205, 27, 8,
  275, 119, 24,
  275, 80,
  215, 126,
  138, 32,
  275, 127
)

yds <- c(
  167, 190, 36,
  125, 76, 19,
  276, 237, 44,
  280, 34,
  147,
  289, 97, 158, 7,
  241, 126, 6,
  268, 69, 44, 25,
  181, 11,

  313, 51, 11,
  147, 12,
  280, 64, 26,
  270, 210, 34, 8,
  281, 141, 29,
  269, 65,
  216, 125,
  132, 23,
  322, 122
)

lie_type <- c(
  'tee', 'fairway', 'rough',
  'tee', 'rough', 'rough',

```

```

'tee', 'fairway', 'rough',
'tee', 'rough',
'tee',
'tee', 'fairway', 'fairway', 'fairway',
'tee', 'fairway', 'fairway',
'tee', 'rough', 'fairway', 'fairway',
'tee', 'rough',

'tee', 'rough', 'rough',
'tee', 'sand',
'tee', 'fairway', 'fairway',
'tee', 'rough', 'rough', 'fairway',
'tee', 'rough', 'rough',
'tee', 'fairway',
'tee', 'fairway',
'tee', 'rough',
'tee', 'rough'
)

target_status <- c(
  'no', 'no', 'yes',
  'no', 'no', 'yes',
  'yes', 'no', 'yes',
  'no', 'yes',
  'yes',
  'yes', 'no', 'no', 'yes',
  'no', 'no', 'yes',
  'no', 'yes', 'no', 'yes',
  'no', 'yes',

  'no', 'no', 'yes',
  'no', 'yes',
  'yes', 'no', 'yes',
  'no', 'no', 'no', 'yes',
  'no', 'no', 'yes',
  'yes', 'yes',
  'yes', 'yes',
  'no', 'yes',
  'no', 'yes'
)

location <- c(
  'short', 'right', 'on_target',
  'right', 'long', 'on_target',
  'on_target', 'right', 'on_target',
  'right', 'on_target',
  'on_target',
  'on_target', 'short', 'long', 'on_target',
  'right', 'short', 'on_target',
  'left', 'on_target', 'short', 'on_target',
  'right', 'on_target',

  'left', 'short', 'on_target',

```

```

'right', 'on_target',
'on_target', 'short', 'on_target',
'right', 'right', 'long', 'on_target',
'right', 'long', 'on_target',
'on_target', 'on_target',
'on_target', 'on_target',
'left', 'on_target',
'right', 'on_target'
)

type_of_shot <- c(
'tee', 'full', 'chip',
'tee', 'chip', 'chip',
'tee', 'full', 'chip',
'tee', 'chip',
'tee',
'tee', 'full', 'full', 'chip',
'tee', 'full', 'chip',
'tee', 'punch', 'chip', 'chip',
'tee', 'chip',

'tee', 'chip', 'chip',
'tee', 'gsbunker',
'tee', 'choked', 'chip',
'tee', 'full', 'chip', 'chip',
'tee', 'full', 'chip',
'tee', 'choked',
'tee', 'full',
'tee', 'chip',
'tee', 'full'
)

```

## Get Shot Metrics

```

con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
# the new annotations (expands from 18 rows by the number determined from the
# get_tracked_shots_data_shape function)

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY date")) %>%
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
  length(club_choice)
  club_metrics <- golf::harmonize_club_metrics(club_metrics = club_metrics_df,
                                              club_choice = club_choice,
                                              distance_to_target = dist_to_target,

```

```

        distance_traveled = yds,
        lie_type = lie_type,
        target_status = target_status,
        location = location,
        type_of_shot = type_of_shot
      )

club_metrics <- club_metrics |>
  dplyr::mutate(dplyr::across(c(dplyr::contains("yds")), ~as.numeric(.x)))
head(club_metrics)
}

club_metrics_df |> dplyr::select(tracked_shots) |> dplyr::mutate(sum(tracked_shots))

con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
# the new annotations (expands from 18 rows by the number determined from the
# get_tracked_shots_data_shape function)

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY date"))
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
  length(club_choice)
  club_metrics <- golf::harmonize_club_metrics(club_metrics = club_metrics_df,
    club_choice = club_choice,
    distance_to_target = dist_to_target,
    distance_traveled = yds,
    lie_type = lie_type,
    target_status = target_status,
    location = location,
    type_of_shot = type_of_shot
  )

  club_metrics <- club_metrics |>
    dplyr::mutate(dplyr::across(c(dplyr::contains("yds")), ~as.numeric(.x)))
  head(club_metrics)
}

club_metrics_df |> dplyr::select(tracked_shots) |> dplyr::mutate(sum(tracked_shots))

```

## Update the Club Metrics Table

```

con <- golf::get_db_connection()

# if the data is newer than the last round in the 'club_metrics' table, append
# the new data to the table

```



```

if ( DBI::dbGetQuery(conn = con, statement = paste0("SELECT DISTINCT date FROM club_metrics ORDER BY da
  dplyr::distinct(date) |>
  unlist() %>%
  lubridate::as_date(.) |>
  as.character() < round_date &

  length(hole_scores) > 0
) {
  DBI::dbAppendTable(conn = con,
    name = 'club_metrics',
    value = club_metrics |>
      dplyr::mutate(date = as.character(date))
  )
}

```

```

# clean the global environment
rm(list = ls()[which(grepl(ls(), pattern= 'con|round_course|round_date|index|real_db')==F)])

```

## Summarize Metrics

### Gather and Format

Gather and format from the database

```

con <- golf::get_db_connection()

# get the hole-by-hole scores from each round in the database

# join the hole-by-hole scores from the 'rounds' table to the 'courses' table
# need each hole's par rating
# need each course's course_rating from the 'courses' table

# join the 'rounds' and 'courses' tables to the 'players' table
# need handicap_index from the 'players' table

scores <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT r.*, c.par, c.course_rating FROM rounds r
  INNER JOIN courses c
  ON c.tees = r.tees
  AND c.course_name = r.course_name
  AND c.hole = r.hole
  INNER JOIN players p
  ON r.GHIN = p.GHIN
  AND r.handicap_index = p.handicap_index
  AND r.date = p.date;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date)) |> # convert strings to date's
  dplyr::relocate(par, .after = hole) |>
  dplyr::relocate(course_rating, .after = tees) |>
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole) |>
  dplyr::ungroup()

```

```

con <- golf::get_db_connection()

# get stroke metrics from the 'club_metrics' table

stroke_quality <- DBI::dbGetQuery(conn = con, statement = paste0(
  "SELECT DISTINCT * FROM club_metrics;"
)) |>
  dplyr::mutate(date = lubridate::as_date(date)) |> # convert strings to date's
  dplyr::group_by(date) |>
  dplyr::arrange(desc(date), hole, stroke) |>
  dplyr::ungroup()

```

## Compute Metrics

Compute standard metrics

```

# define metrics
scores_sum <- scores |>
  dplyr::mutate(date_course = paste0(date,
                                     '\n',
                                     course_name, '\n',
                                     handicap_index),
    chips = dplyr::case_when(
      grepl(date, pattern = '07-13') ~ NA_real_, TRUE ~ chips),
    `chips+putts` = chips+putts,
    FIR_opps = dplyr::case_when(par > 3 ~ 1, TRUE ~ NA),
    `Iron FIRs` = dplyr::case_when(par > 3 &
      grepl(tee_club, pattern = '4|5') &
      FIR == 1 ~ 1,
      TRUE ~ 0),
    `Iron FIR opps` = dplyr::case_when(par > 3 &
      grepl(tee_club, pattern = '4|5') ~ 1,
      TRUE ~ 0),
    `Driver FIRs` = dplyr::case_when(par > 3 &
      tee_club == 'D' &
      FIR == 1 ~ 1,
      TRUE ~ 0),
    `Driver FIR opps` = dplyr::case_when(par > 3 &
      tee_club == 'D' ~ 1,
      TRUE ~ 0),
    `Par 3 GIRs` = dplyr::case_when(par == 3 &
      GIR == 1 ~ 1,
      TRUE ~ 0),
    greenie_putts = dplyr::case_when(GIR == 1 ~ putts, TRUE ~ NA_real_),
    updown_conv = dplyr::case_when(GIR == 0 &
      par == gross ~ 1,
      TRUE ~ 0),
    updown_opps = dplyr::case_when(GIR == 0 &
      chips > 0 ~ 1,
      TRUE ~ 0)
  ) |>
  dplyr::rename(`Handicap Index` = handicap_index)

```

```

# summarize by round
scores_sum <- scores_sum |>
  dplyr::group_by(date, date_course, course_rating, `Handicap Index`) |>
  dplyr::summarize(
    FIRs = sum(FIR, na.rm = F),
    `Iron FIRs` = sum(`Iron FIRs`, na.rm = T),
    `Iron FIR%` = round(((sum(`Iron FIRs`, na.rm = T)/sum(`Iron FIR opps`, na.rm = T))*100, 1),
    `Driver FIRs` = sum(`Driver FIRs`, na.rm = T),
    `Driver FIR%` = round(((sum(`Driver FIRs`, na.rm = T)/sum(`Driver FIR opps`, na.rm = T))*100, 1),
    `FIR%` = round((sum(FIRs, na.rm = T)/sum(FIR_opps, na.rm = T))*100, 1),
    GIRs = sum(GIR, na.rm = F),
    `Par 3 GIRs` = sum(`Par 3 GIRs`, na.rm = T),
    `GIR%` = round((sum(GIRs, na.rm = T)/18)*100, 1),
    putts = sum(putts, na.rm = F),
    `Avg GIR putts` = round(mean(greenie_putts, na.rm = T), 2),
    chips = sum(chips, na.rm = F),
    `chips+putts` = sum(`chips+putts`, na.rm = F),
    `UpDown%` = round(((sum(updown_conv, na.rm = T)/sum(updown_opps, na.rm = T))*100), 1),
    pars = sum(is_gross_par, na.rm = F),
    birdies = sum(is_gross_birdie, na.rm = F),
    bogies = sum(is_gross_bogey, na.rm = F),
    `doubles+` = sum(is_gross_bogey_worse, na.rm = F),
    penalties = sum(penalties, na.rm = T),
    `Gross Score` = mean(tot_gross, na.rm = F),
    `Net Score` = mean(tot_net, na.rm = F)) %>%
  dplyr::mutate(dplyr::across(c(`Iron FIRs`, `Driver FIRs`,
    `Iron FIR%`, `Driver FIR%`, `FIR%`),
    ~dplyr::if_else(is.na(FIRs), NA, .)),

    dplyr::across(c(`Par 3 GIRs`, `Avg GIR putts`,
    `UpDown%`, `GIR%`),
    ~dplyr::if_else(is.na(GIRs), NA, .)),

    `chips+putts` = dplyr::case_when(is.na(chips) ~ NA_real_,
    TRUE ~ `chips+putts`)) %>%

  dplyr::mutate(
    `UpAndDown%` = dplyr::case_when(
      grepl(date, pattern = '07-13|09-21') ~ NA, TRUE ~ `UpDown%`),

    `Iron FIR%` = dplyr::case_when(`Iron FIR%` == NaN ~ 0.0, TRUE ~ `Iron FIR%`))

head(scores_sum |>
  dplyr::arrange(desc(date)))

```

```

## # A tibble: 6 x 26
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `Handicap Index`  FIRs `Iron FIRs` `Iron FIR%` `Driver
##   <date>      <chr>              <dbl>          <dbl> <int>      <dbl>      <dbl>
## 1 2026-08-16 "2026-08-16\nDell~      68.5          9.9      6         1        100
## 2 2026-08-02 "2026-08-02\nRand~      69.8          9.8      2         0         NaN

```

```
## 3 2026-07-31 "2026-07-31\nSilv~          68.9          9.8          5          0          NaN
## 4 2026-07-26 "2026-07-26\nRand~          69.8         10.2          3          0          NaN
## 5 2026-07-12 "2026-07-12\nDell~          67.8         10.2          5          1         100
## 6 2026-07-05 "2026-07-05\nRand~          69.8         10.5          9          0          NaN
## # i 9 more variables: `UpDown%` <dbl>, pars <int>, birdies <int>, bogies <int>, `doubles+` <int>, per
```

## View Metrics

Separate and view metrics

## Scoring Metrics Scores and Handicap

```
scoring_metrics <- scores_sum |>
  dplyr::select(`Handicap Index`, `Gross Score`, `Net Score`)
head(scoring_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 6
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `Handicap Index` `Gross Score` `Net Sc
##   <date>    <chr>                <dbl>          <dbl>          <dbl>      <
## 1 2026-08-16 "2026-08-16\nDell Urich\n9.9"      68.5            9.9            78
## 2 2026-08-02 "2026-08-02\nRandolph North\n9.8"      69.8            9.8            82
## 3 2026-07-31 "2026-07-31\nSilverbell\n9.8"      68.9            9.8            82
## 4 2026-07-26 "2026-07-26\nRandolph North\n10.2"      69.8           10.2            80
## 5 2026-07-12 "2026-07-12\nDell Urich\n10.2"      67.8           10.2            85
## 6 2026-07-05 "2026-07-05\nRandolph North\n10.5"      69.8           10.5            78
```

**Stroke Metrics** In golf, every hole has a **par**— an average number of strokes taken to get the ball in the hole

There are (almost always), three different **pars** on every course:

- par 3s
- par 4s
- par 5s

A shorthand to determine how a golfer performed across holes is to rate how many strokes *relative to par* they were (-1, -2, +1, +2, etc)

These numbers are given names:

- 0 = par
- -2 = eagle
- -1 = birdie
- +1 = bogey
- +2 = double bogey

I quantify these below:

```
stroke_metrics <- scores_sum |>
  dplyr::select(`doubles+`, bogies, pars, birdies)
head(stroke_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 7
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating `doubles+` bogies  pars birdies
##   <date>    <chr>                <dbl>      <int>  <int> <int>  <int>
## 1 2026-08-16 "2026-08-16\nDell Urich\n9.9"      68.5         2      7     7      1
## 2 2026-08-02 "2026-08-02\nRandolph North\n9.8"    69.8         1      9     7      1
## 3 2026-07-31 "2026-07-31\nSilverbell\n9.8"      68.9         1     10     7      0
## 4 2026-07-26 "2026-07-26\nRandolph North\n10.2"    69.8         2      5     9      2
## 5 2026-07-12 "2026-07-12\nDell Urich\n10.2"    67.8         2     10     6      0
## 6 2026-07-05 "2026-07-05\nRandolph North\n10.5"    69.8         2      4    10      2
```

**Around-the-Green Metrics** Chips: + strokes around the green taken to get onto the green

Putts: + strokes taken with the putter on the green

Avg GIR putts: + Average # of putts on holes where the ball was hit on to the green within 2 strokes of par (green in regulation, GIR)

UpDown% (aka Scramble%): + # of holes without a GIR but par was made / # of holes without a GIR

```
atg_metrics <- scores_sum |>
  dplyr::select(chips, `chips+putts`, `UpAndDown%`, putts, `Avg GIR putts`)
head(atg_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 8
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating chips `chips+putts` `UpAndDown%` putts
##   <date>    <chr>                <dbl>  <dbl>      <dbl>      <dbl> <int>
## 1 2026-08-16 "2026-08-16\nDell Urich\n9.9"      68.5    16         46      33.3    30
## 2 2026-08-02 "2026-08-02\nRandolph North\n9.8"    69.8    14         46      20      32
## 3 2026-07-31 "2026-07-31\nSilverbell\n9.8"      68.9    18         49      21.4    31
## 4 2026-07-26 "2026-07-26\nRandolph North\n10.2"    69.8    17         47      40      30
## 5 2026-07-12 "2026-07-12\nDell Urich\n10.2"    67.8    15         52      20      37
## 6 2026-07-05 "2026-07-05\nRandolph North\n10.5"    69.8    12         42      62.5    30
```

**Ball Striking Metrics** Approach and tee accuracy

GIR (green in regulation): + hole where the ball was hit on to the green within 2 strokes of par

FIR (fairway in regulation): + hole where the ball was hit on to the fairway from the tee box (only on par 4's and par 5's)

Iron FIR: + hole where an iron was used off the tee for a FIR

Driver FIR: + hole where driver was used off the tee for a FIR

```
ball_striking_metrics <- scores_sum |>
  dplyr::select(GIRs, `GIR%`, `Par 3 GIRs`,
               FIRs, `FIR%`, `Iron FIRs`, `Iron FIR%`,
               `Driver FIRs`, `Driver FIR%`)
head(ball_striking_metrics |>
  dplyr::arrange(desc(date)))
```

```
## # A tibble: 6 x 12
## # Groups:   date, date_course, course_rating [6]
##   date      date_course      course_rating  GIRs `GIR%` `Par 3 GIRs`  FIRs `FIR%`
##   <date>    <chr>          <dbl> <int> <dbl>      <dbl> <int> <dbl>
## 1 2026-08-16 "2026-08-16\nDell Urich\n9.9"      68.5     6  33.3         1     6  46.2
## 2 2026-08-02 "2026-08-02\nRandolph North\n9.8"    69.8     8  44.4         2     2  14.3
## 3 2026-07-31 "2026-07-31\nSilverbell\n9.8"    68.9     4  22.2         2     5  38.5
## 4 2026-07-26 "2026-07-26\nRandolph North\n10.2"  69.8     7  38.9         1     3  21.4
## 5 2026-07-12 "2026-07-12\nDell Urich\n10.2"    67.8     6  33.3         2     5  38.5
## 6 2026-07-05 "2026-07-05\nRandolph North\n10.5"    69.8     9   50         1     9  64.3
```

**Club Metrics** Yardage and accuracy for each club

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date      club      n rd_min_yds_to_target rd_min_yds_traveled rd_min_yd_diff
##   <date>    <chr> <int>          <dbl>          <dbl>          <dbl>
## 1 2026-08-16 3W         3          253             97            -13
## 2 2026-08-16 4         1          215            216             -1
## 3 2026-08-16 5         1          205            210             -5
## 4 2026-08-16 6         1          192            190              2
## 5 2026-08-16 7         1           70             69              1
## 6 2026-08-16 8         2          161            125             -2
```

```
## # A tibble: 6 x 6
## # Groups:   date [1]
##   date      club      n rd_max_yds_to_target rd_max_yds_traveled rd_max_yd_diff
##   <date>    <chr> <int>          <dbl>          <dbl>          <dbl>
## 1 2026-08-16 3W         3          266            268            156
## 2 2026-08-16 4         1          215            216             -1
## 3 2026-08-16 5         1          205            210             -5
## 4 2026-08-16 6         1          192            190              2
## 5 2026-08-16 7         1           70             69              1
## 6 2026-08-16 8         2          179            181             36
```

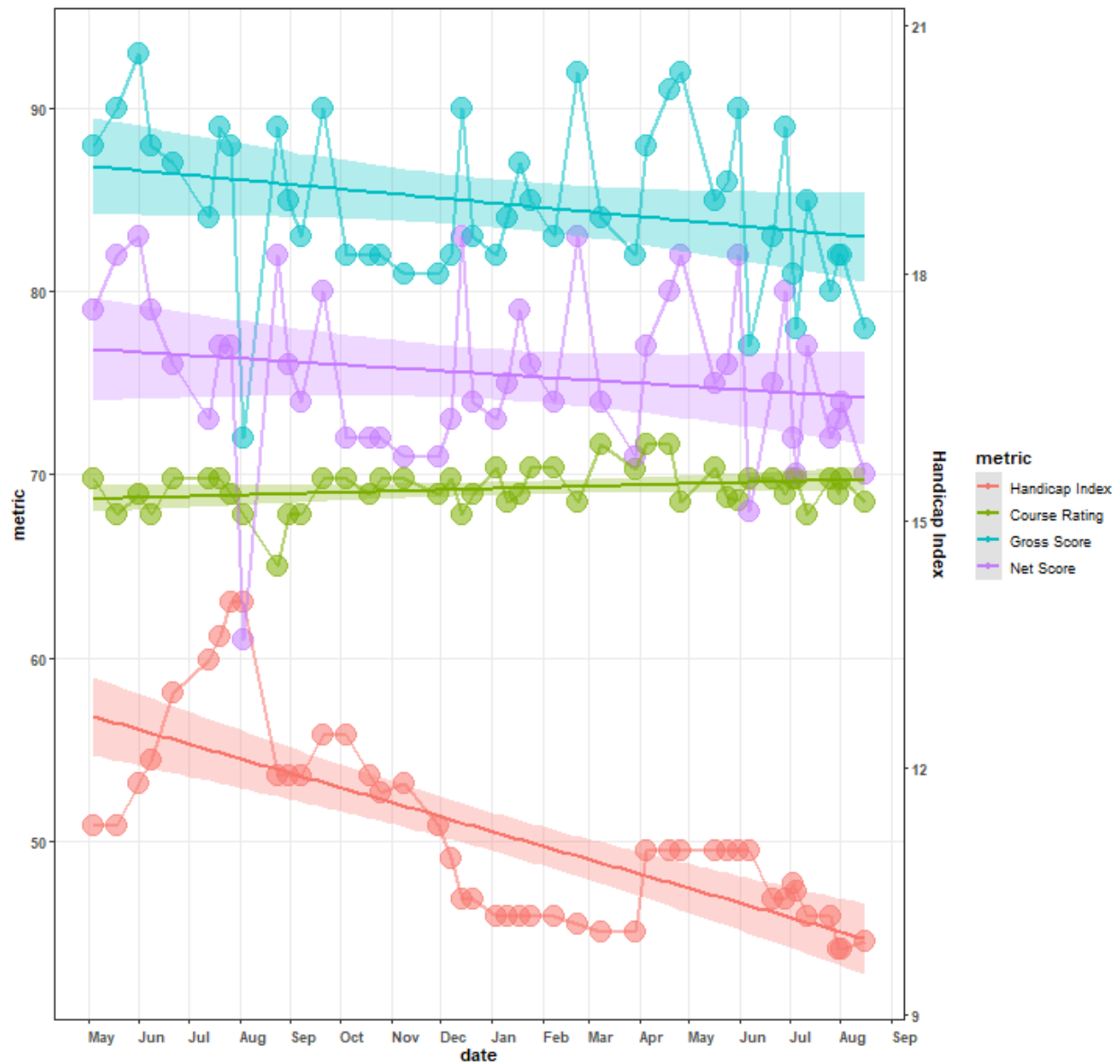
```
## # A tibble: 6 x 10
## # Groups:   date [1]
##   date      club `rd club strokes` miss_direction `rd club miss dir` rd_avg_yds_to_target rd_avg_yd_diff
##   <date>    <chr>          <int> <chr>          <int>          <dbl>
## 1 2026-08-16 3W         3 left          1          258
## 2 2026-08-16 3W         3 right         1          258
## 3 2026-08-16 3W         3 short         1          258
## 4 2026-08-16 4         1 on_target     1          215
## 5 2026-08-16 5         1 right         1          205
## 6 2026-08-16 6         1 right         1          192
```

## Plot Metric Summaries

### Scoring Metrics

```
scoring_metrics |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
    `Handicap Index` = `Handicap Index`*4.5) |>
  dplyr::rename(`Course Rating` = course_rating) |>
  dplyr::group_by(date, date_course, `Handicap Index`) |>
  tidyr::pivot_longer(cols = c(`Course Rating`, `Net Score`), names_to = 'metric', values_to = 'value', 'I
  ggplot2::ggplot(aes(x = date_course, y = value)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1,
    color = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
    fill = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1,
    color = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
    fill = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
  ggplot2::geom_smooth(aes(x = date, y = value, group = metric,
    color = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
    fill = factor(metric,
      levels = c('Handicap Index', 'Course Rating', 'Gross Score', 'I
    alpha = 0.3, method = 'lm') +
  ggplot2::scale_y_continuous(sec.axis = ggplot2::sec_axis(~./4.5, name = 'Handicap Index'))+
  ggplot2::labs(title = paste0('Performance Over Time\n',
    scores |>
      dplyr::distinct(date) |>
      dplyr::last() |>
      unlist() %>% lubridate::as_date(.) |>
      as.character(),
      ' - ',
    scores |>
      dplyr::distinct(date) |>
      dplyr::first() |>
      unlist() %>% lubridate::as_date(.) |>
      as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric'
  ) +
  ggplot2::guides(alpha = 'none', size = 'none', fill = 'none') +
  ggplot2::theme_bw() +
  ggplot2::theme(panel.grid.minor = ggplot2::element_blank(),
    axis.title = ggplot2::element_text(face = 'bold'),
    axis.text.y = ggplot2::element_text(face = 'bold'),
    axis.text.x = ggplot2::element_text(face = 'bold', angle = 0, hjust = 0, vjust = 0),
    title = ggplot2::element_text(face = 'bold')
  ) +
  ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b')
```

## Performance Over Time 2025-05-04 - 2026-08-16



## Stroke Metrics

```
stroke_metrics |>
  tidyr::pivot_longer(cols = c(`doubles+`:birdies), names_to = 'metric', values_to = 'value', values_drop_na = T) +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Hole-by-Hole Scores Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

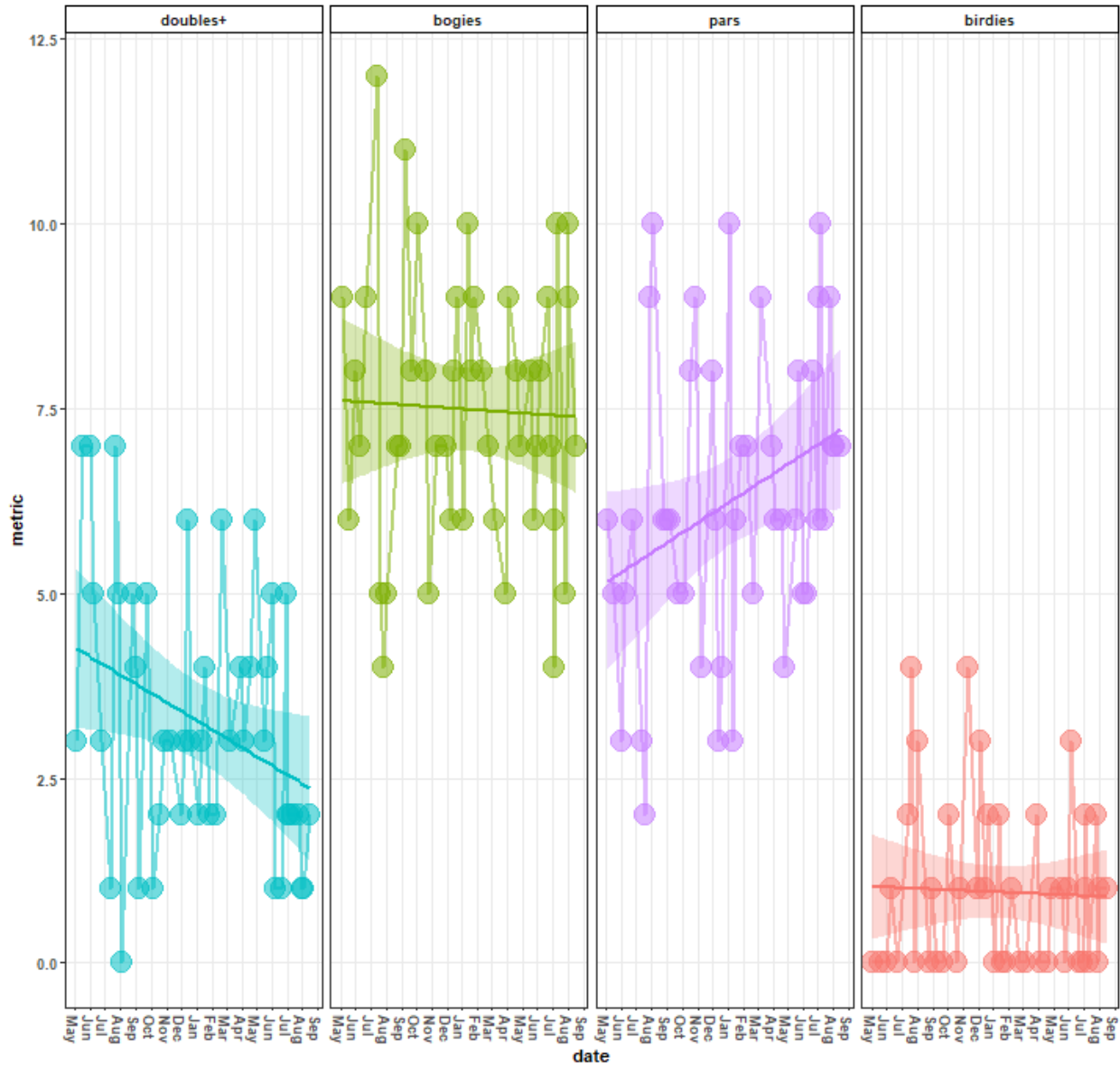


```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('doubles+', 'bogies', 'pars', 'birdies')), nrow = 1)

```

### Hole-by-Hole Scores Over Time 2025-05-04 - 2026-08-16



### Around the Green Metrics

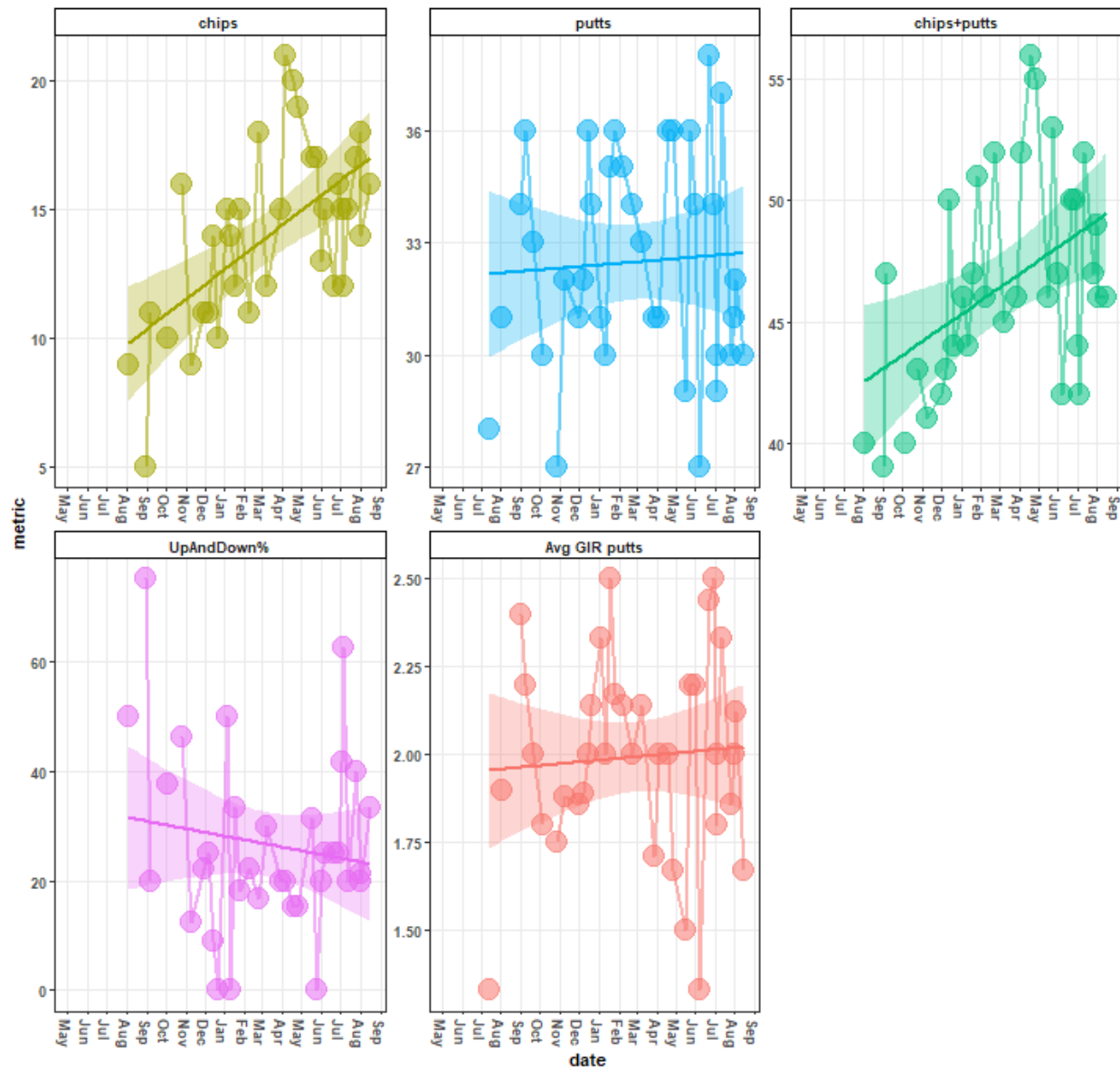
```
atg_metrics |>
  tidyr::pivot_longer(cols = c(chips:`Avg GIR putts`), names_to = 'metric', values_to = 'value', values_from = 'score') +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Around the Green Metrics Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
    panel.grid.minor = ggplot2::element_blank(),
    axis.title = ggplot2::element_text(face = 'bold'),
    axis.text.y = ggplot2::element_text(face = 'bold'),
    axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
    title = ggplot2::element_text(face = 'bold'),
    strip.text.x.top = ggplot2::element_text(face = 'bold'),
    strip.background = ggplot2::element_rect(color = 'black',
        fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('chips', 'putts', 'chips+putts', 'UpAndDown%', 'Avg GII

```

## Around the Green Metrics Over Time 2025-05-04 - 2026-08-16



## Ball Striking Metrics

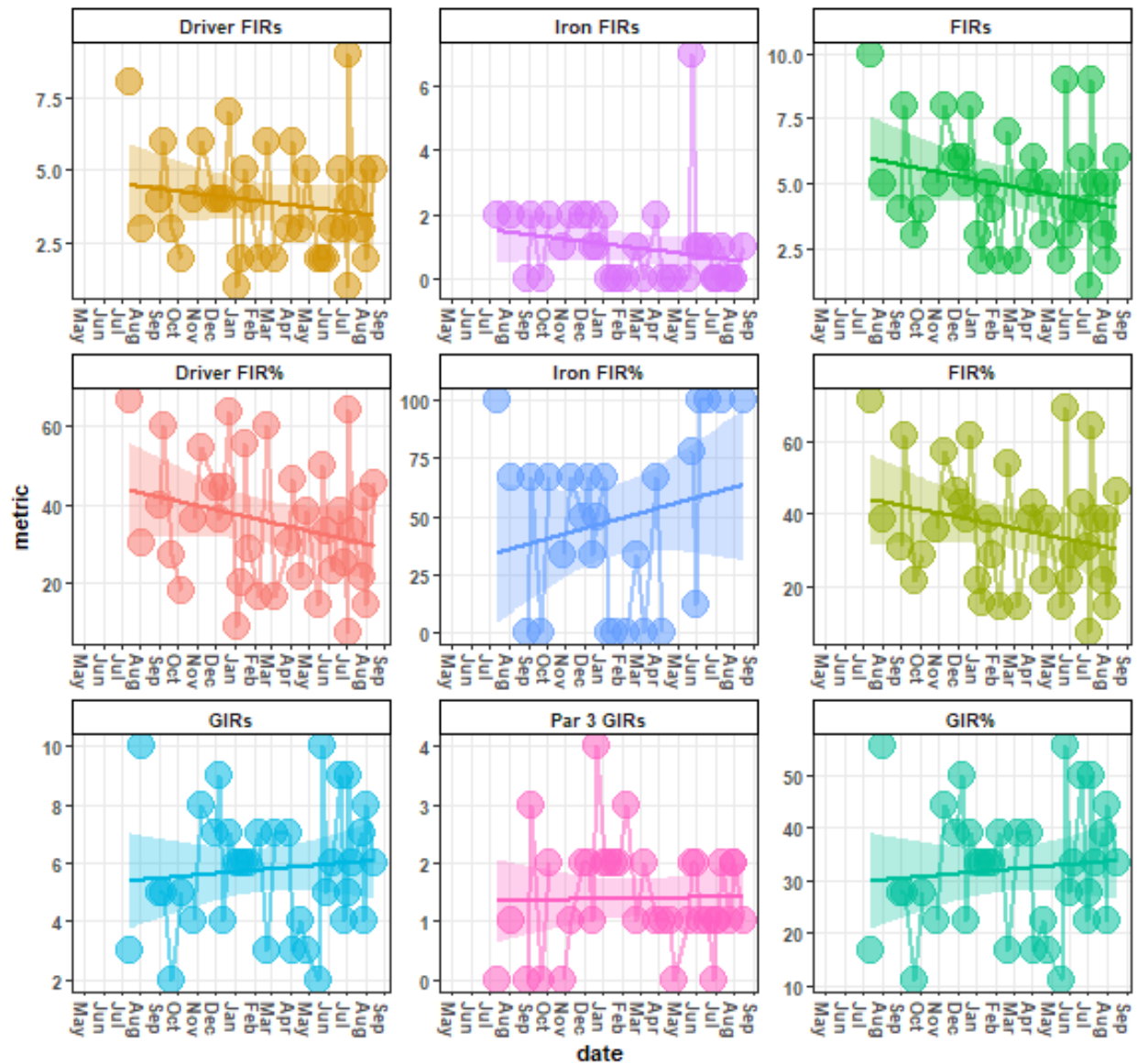
```
ball_striking_metrics |>
  tidyr::pivot_longer(cols = c(GIRs:`Driver FIR%`), names_to = 'metric', values_to = 'value', values_drop_na = T) +
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Ball Striking Over Time\n',
                                scores |>
                                  dplyr::distinct(date) |>
                                    dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
               panel.grid.minor = ggplot2::element_blank(),
               axis.title = ggplot2::element_text(face = 'bold'),
               axis.text.y = ggplot2::element_text(face = 'bold'),
               axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
               title = ggplot2::element_text(face = 'bold'),
               strip.text.x.top = ggplot2::element_text(face = 'bold'),
               strip.background = ggplot2::element_rect(color = 'black',
                                                         fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric, levels = c('Driver FIRs', 'Iron FIRs', 'FIRs',
                                              'Driver FIR%', 'Iron FIR%', 'FIR%',
                                              'GIRs', 'Par 3 GIRs', 'GIR%')), scales = 'free')

```

## Ball Striking Over Time 2025-05-04 - 2026-08-16



## Stroke Quality Metrics

```
stroke_quality_min |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(
    grepl(metric, pattern = 'target') ~ 'min. target distance',
    grepl(metric, pattern = 'traveled') ~ 'min. shot distance',
    grepl(metric, pattern = 'diff') ~ 'min. \ntarget dist. - shot d

  dplyr::mutate(date = lubridate::as_date(date),
    club = as.character(club),
```

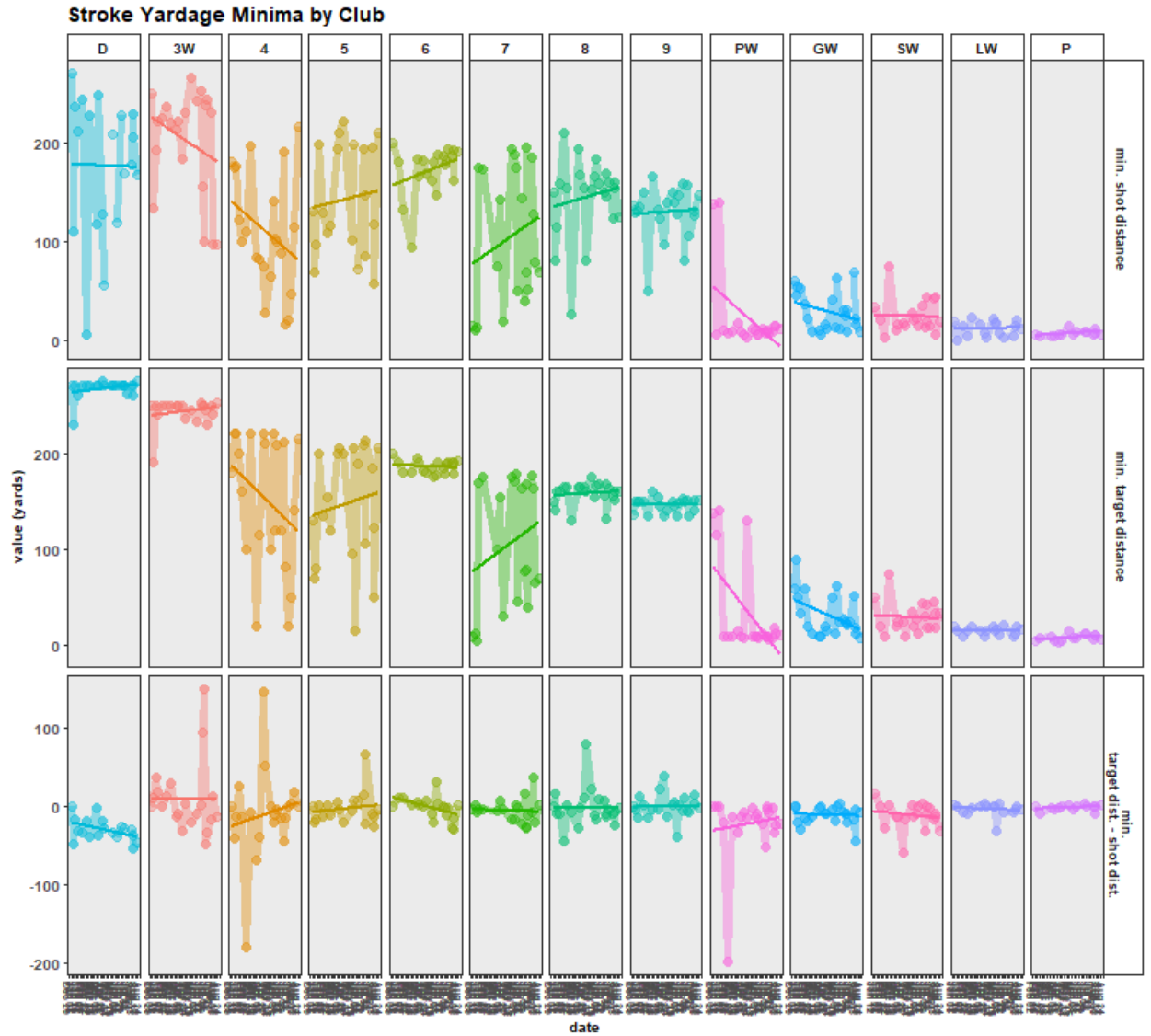
```

      n = as.integer(n)) |>
ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
ggplot2::geom_point(alpha = 0.4, size = 3) +
ggplot2::geom_line(alpha = 0.4, size = 2) +
ggplot2::geom_smooth(method = 'lm', se = F) +
ggplot2::labs(title = 'Stroke Yardage Minima by Club', x = 'date', y = 'value (yards)', fill = 'club')
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                     '6', '7', '8', '9',
                                     'PW', 'GW', 'SW', 'LW', 'P'))),
          scales = 'free')

```



Minima

```
stroke_quality_max |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(c(date:n), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains('yd')), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'max. target distance',
                                          grepl(metric, pattern = 'traveled') ~ 'max. shot distance',
                                          grepl(metric, pattern = 'diff') ~ 'max.\ntarget dist. - shot dist.'))
  dplyr::mutate(date = lubridate::as_date(date),
                club = as.character(club),
                n = as.integer(n)) |>
  ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
  ggplot2::geom_point(alpha = 0.4, size = 3) +
  ggplot2::geom_line(alpha = 0.4, size = 2) +
  ggplot2::geom_smooth(method = 'lm', se = F) +
  ggplot2::labs(title = 'Stroke Yardage Maxima by Club', x = 'date', y = 'value (yards)', fill = 'club')
```



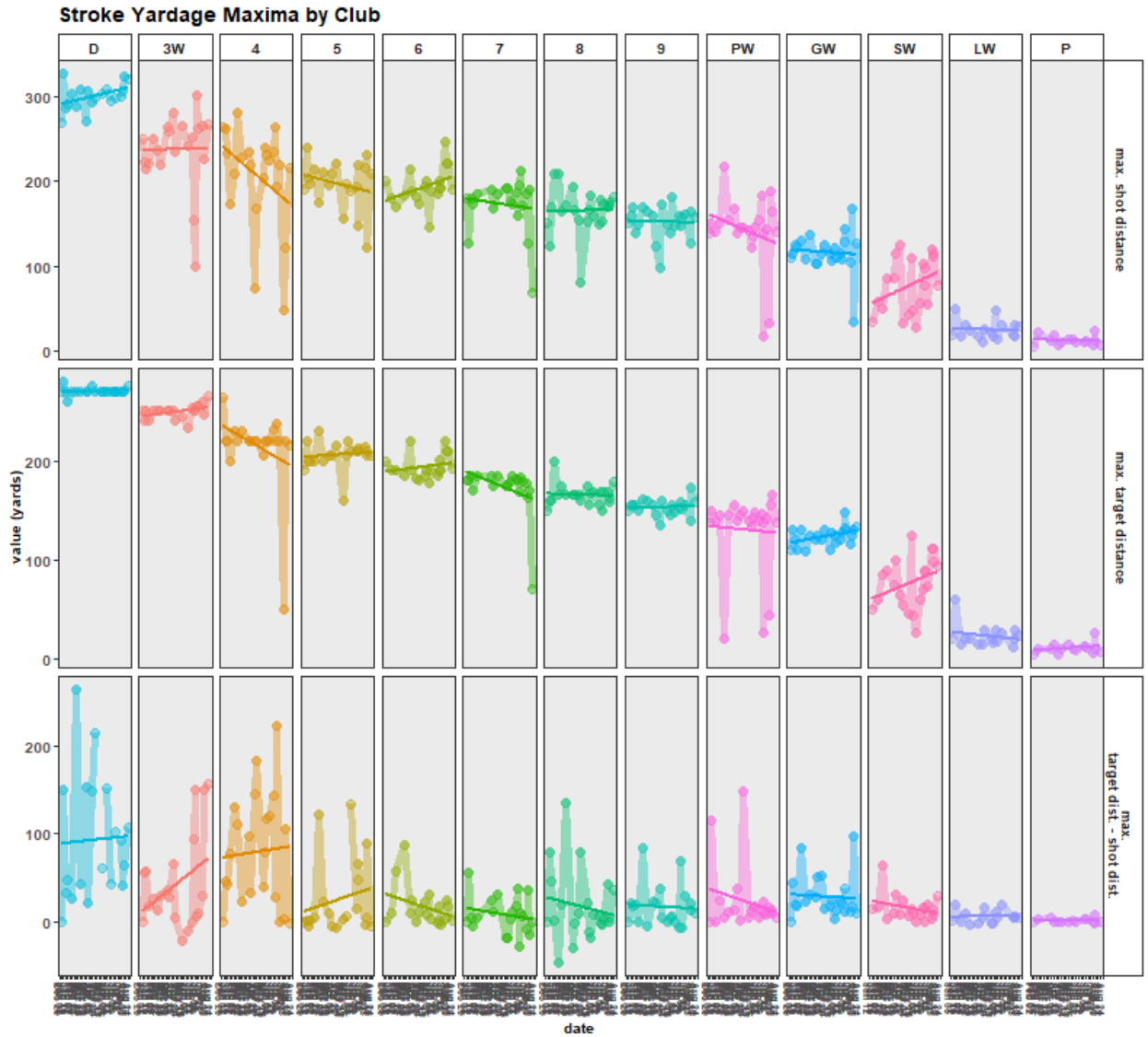
```

ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(title = ggplot2::element_text(face = 'bold', size = 12),
               axis.title = ggplot2::element_text(face = 'bold', size = 10),
               axis.text = ggplot2::element_text(face = 'bold', size = 10),
               axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
               legend.position = 'none',
               strip.background = ggplot2::element_rect(fill = 'white'),
               strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
               strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                     '6', '7', '8', '9',
                                     'PW', 'GW', 'SW', 'LW', 'P'))),
           scales = 'free')

```



Maxima

```
full_stroke_quality_avg |>
  dplyr::ungroup() %>%
  dplyr::mutate(dplyr::across(dplyr::everything(), ~as.character(.x))) |>
  tidyr::pivot_longer(cols = c(dplyr::contains("rd")), names_to = 'metric', values_to = 'vals') |>
  dplyr::mutate(metric = dplyr::case_when(grepl(metric, pattern = 'target') ~ 'avg. target distance',
    grepl(metric, pattern = 'traveled') ~ 'avg. shot distance',
    grepl(metric, pattern = 'diff') ~ 'avg. target dist. - shot dist.',
    # grepl(metric, pattern = 'accuracy') ~ 'avg. accuracy',
    grepl(metric, pattern = 'strokes') ~ 'n strokes',
    grepl(metric, pattern = 'dir') ~ 'strokes w/club\nin round\nby',
    grepl(metric, pattern = 'acc') ~ '% accuracy')) |>

  dplyr::filter(!grepl(metric, pattern = 'strokes')) |>
  dplyr::mutate(date = lubridate::as_date(date),
    club = as.character(club),
    vals = as.numeric(vals))
```

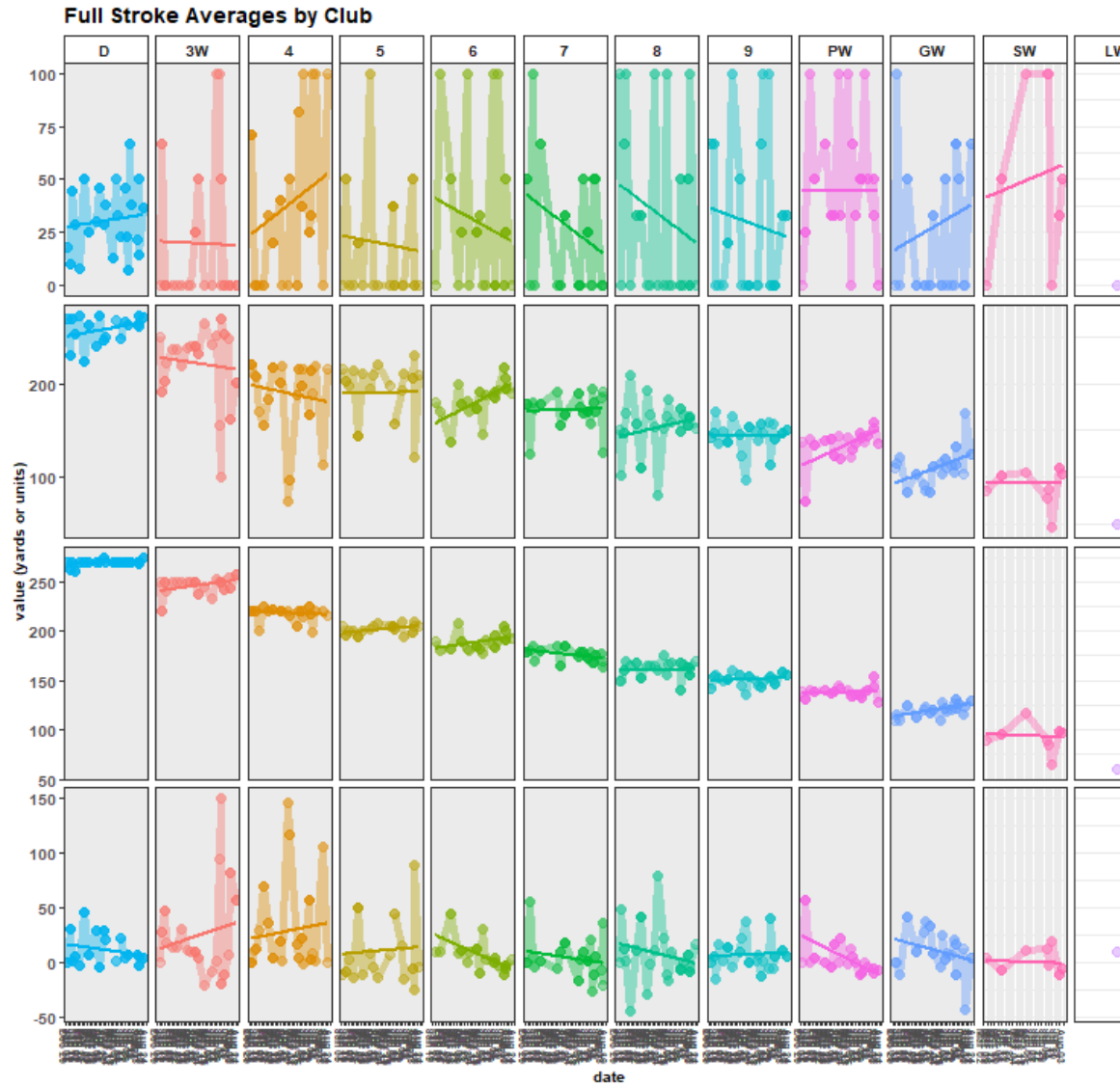
```

    ) |>
ggplot2::ggplot(aes(x = date, y = vals, group = metric, color = club, fill = club)) +
ggplot2::geom_point(alpha = 0.4, size = 3) +
ggplot2::geom_line(alpha = 0.4, size = 2) +
ggplot2::geom_smooth(method = 'lm', se = F) +
ggplot2::labs(title = 'Full Stroke Averages by Club', x = 'date', y = 'value (yards or units)', fill = club) +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(
  title = ggplot2::element_text(face = 'bold', size = 12),
  axis.title = ggplot2::element_text(face = 'bold', size = 10),
  axis.text = ggplot2::element_text(face = 'bold', size = 10),
  axis.text.x = ggplot2::element_text(face = 'bold', size = 7, angle = 270, hjust = 0, vjust = 1),
  legend.position = 'none',
  strip.background = ggplot2::element_rect(fill = 'white'),
  strip.text.y.right = ggplot2::element_text(face = 'bold', size = 9),
  strip.text.x.top = ggplot2::element_text(face = 'bold', size = 10)) +
ggplot2::scale_x_date(date_breaks = 'weeks', date_labels = '%b %d') +
ggplot2::facet_grid(

  rows = vars(metric),

  cols = vars(factor(club, levels = c('D', '3W', '4', '5',
                                     '6', '7', '8', '9',
                                     'PW', 'GW', 'SW', 'LW', 'P'))),
          scales = 'free')

```



Full Stroke Average

Main Metrics

```
scores_sum |>
  dplyr::mutate(course_rating = dplyr::case_when(grepl(date_course, pattern = 'Sewailo') ~ 68.9, TRUE ~
  tidyr::pivot_longer(cols = c(`Handicap Index`, `Gross Score`, `Net Score`,
    `doubles+`, bogies, pars, birdies, chips, putts,
    `UpDown%`, `Avg GIR putts`, `GIR%`, `Par 3 GIRs`,
    `Iron FIR%`, `Driver FIR%`), names_to = 'metric', values_to = 'value', v
  ggplot2::ggplot(aes(x = date_course, y = value, group = metric, color = metric, fill = metric)) +
  ggplot2::geom_point(aes(x = date, y = value, size = 4, alpha = 0.1), na.rm = T) +
  ggplot2::geom_line(aes(x = date, y = value, size = 1, alpha = 0.1), na.rm = T) +
  ggplot2::geom_smooth(aes(x = date, y = value), alpha = 0.3, method = 'lm') +
  ggplot2::labs(title = paste0('Performance Over Time\n',
    scores |>
      dplyr::distinct(date) |>
      dplyr::last() |>
```

```

        unlist() %>% lubridate::as_date(.) |>
        as.character(),
        ' - ',
        scores |>
        dplyr::distinct(date) |>
        dplyr::first() |>
        unlist() %>% lubridate::as_date(.) |>
        as.character()),
    x = 'date',
    y = 'metric',
    color = 'metric', fill = 'metric') +
ggplot2::guides(alpha = 'none', size = 'none') +
ggplot2::theme_bw() +
ggplot2::theme(legend.position = 'none',
    panel.grid.minor = ggplot2::element_blank(),
    axis.title = ggplot2::element_text(face = 'bold'),
    axis.text.y = ggplot2::element_text(face = 'bold'),
    axis.text.x = ggplot2::element_text(face = 'bold', angle = 270, hjust = 0, vjust = 0.5),
    title = ggplot2::element_text(face = 'bold'),
    strip.text.x.top = ggplot2::element_text(face = 'bold'),
    strip.background = ggplot2::element_rect(color = 'black',
        fill = 'white')) +
ggplot2::scale_x_date(date_breaks = '1 month', date_labels = '%b') +
ggplot2::facet_wrap(~factor(metric,
    levels = c('Handicap Index', 'Gross Score', 'Net Score',
        'doubles+', 'bogies', 'pars', 'birdies',
        'chips', 'putts', 'UpDown%', 'Avg GIR putts',
        'GIR%', 'Par 3 GIRs', 'Iron FIR%', 'Driver FIR%')),
    scales = 'free',
    ncol = 4)

```

# Performance Over Time 2025-05-04 - 2026-08-16

